

STM32MP1 Series

Microprocessors



Multicore STM32MP1 architecture is ideal for Open Source Linux based applications with real-time and power constrained subsystems

The STM32MP1 series embed a dual Arm® Cortex®-A7, Cortex®-M4 and a 3D GPU. This flexible architecture allows high processing and real-time tasks in a single chip. It comes with large packages offer supporting lowest PCB cost structure and smallest board space.

STM32MP1 series is drastically reducing development time thanks to OpenSTLinux Distribution as a Mainlined Open Source Linux Distribution and STM32Cube Tools specially upgraded to cope with Linux MPU development.

STM32MP1 Embedded software distribution includes:

- Linux® distribution based on Yocto, running on the Arm® Cortex®-A processor(s): **OpenSTLinux Distribution**



- STM32Cube MPU Package, running on the Arm® Cortex®-M processor: STM32CubeMP1 Package



STM32MP157A-DK1
STM32MP157C-DK2
2 Discovery Kits



Flash this code to access to our wiki !
<https://wiki.st.com/stm32mpu>



TARGETED APPLICATIONS

- Industrial
- Home
- Consumer
- Health and Wellness

CORE

- Arm® Dual Cortex®-A7 up to @ 800 MHz
- Arm® Cortex®-M4 core @ 209 MHz

EXTERNAL MEMORIES SUPPORT

- DDR3, DDR3L, LPDDR2, LPDDR3
- SLC NAND, SPI NAND
- eMMC, SD card, Quad-SPI NOR

INTERNAL MEMORIES

- System RAM 256kB
- MCU RAM 484kB

ANALOG

- 2x 16-bit ADCs
- 2x 12-bit DACs

GRAPHICS

- 3D GPU OpenGL ES 2.0
- LCD-TFT Controller
- MIPI-DSI 2 lanes

SECURITY

- TrustZone
- AES 256, TDES
- SHA-256, MD5, HMAC
- Secure boot, RAMs & Peripherals

OTHER

- Up to 176 GPIOs
- Up to 125°C supported as maximum junction temperature

Authorized distributors

- Arrow: Anurag: +91 9910645646 • RamaKrishna: Sandeep: +91 9810209232
- Avnet : Shobhan : +91 9227155584 • EDOM: Om Prakash: +91 9910075895
- Future: Sunil: +91 9844640436 • Nexty: Gajraj: +91 9560225544
- Teck: Surendranath: +91 9810728299 • WT: Ramkumar: +91 9008088998
- Yosun: Amit: +91 9902509977

